

Andrews Acheampong

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Kumasi, Ashanti - AK-144-8245, Ghana

RESEARCH INTERESTS

I am interested in scalable Bayesian inference and distributed MCMC, and how such methodologies can be developed and applied to challenging problems in molecular biology and genomics.

EDUCATION

- **Kwame Nkrumah University of Science and Technology**  January 2026 - Present
MPhil. Biodata Analytics and Computational Genomics
Kumasi, Ghana
 - First Semester Results:
 - * WA: 72.33/100
 - * GPA Equivalent: 3.53/4.00
 - * Credits Obtained: 18
- **Kwame Nkrumah University of Science and Technology**  January 2021 - August 2024
Bachelor of Science in Statistics
Kumasi, Ghana
 - CWA: 74.49/100
 - GPA Equivalent: 3.82/4.00
 - Total Credits Obtained: 154

RESEARCH EXPERIENCE AND LABORATORY EXPERIENCE

- **Genetic and Statistical Foundations of Coalescent Theory and Its Applications in Population Genetics**  July 2026
Graduate Term Paper
 - Examined coalescent theory through the Wright-Fisher model, stochastic processes, and backward-time population genetic inference.
 - Applied geometric, exponential, and Poisson models alongside Watterson's estimator, nucleotide diversity, the site-frequency spectrum, and Tajima's (D).
 - Investigated statistical inference of effective population size, demographic history, migration, natural selection, and genetic ancestry.
- **Kumasi Veterinary Laboratory, Molecular Department** May 4-22, 2026
Wet Laboratory Intern, Molecular Diagnostics
Kumasi, Ghana
 - Completed supervised molecular diagnostics training involving DNA and RNA extraction from laboratory samples.
 - Prepared PCR master mixes and carried out PCR-based viral detection workflows under supervision.
 - Interpreted amplification outputs with supervisor guidance, including the expected exponential amplification curve pattern for target-positive reactions.
 - Practiced contamination-aware sample handling, pipetting accuracy, laboratory documentation, and careful reaction setup.
- **Predictive Analytics for Kidney Disease Progression** 
Undergraduate Thesis
 - Investigated the impact of hemoglobin, serum creatinine, hypertension, and random blood glucose on CKD risk.
 - Developed a predictive model for Chronic Kidney Disease (CKD) using logistic regression.
 - Achieved 97% accuracy and a Kappa value of 0.94 through 10-fold cross-validation.

ACADEMIC AND TEACHING APPOINTMENTS

- **Department of Statistics and Actuarial Science, KNUST**  October 2024 - October 2025
Research and Teaching Assistant
Kumasi, Ghana
 - Conduct tutorial sessions for Probability Distributions and assisting undergraduate students in problem-solving.
 - Design and prepare homework activities for students in Probability and Measure Theory, and Probability and Statistical Inference.
 - Assist faculty members in research projects focused on data science and machine learning, providing guidance to final-year undergraduate students.
- **Self-Employed** September 2024 - July 2025
Instructor
Remote
 - Instructing students the basics of data analysis using R.
 - Developed and delivered course materials covering data manipulation, visualization, and statistical analysis in R.
 - Provided hands-on experience with real-world datasets and case studies focused on medical applications.

BIOINFORMATICS AND COMPUTATIONAL PROJECTS

- **Rhinovirus Classification Web App: [Web application]**

January 2025

Tools: [Python, Flask, Jinja2, SQLite, Matplotlib, Biopython, HTML, JavaScript, CSS]



- Developed a web application for analyzing and visualizing FASTA sequence data, supporting Rhinovirus classification.
- Integrated phylogenetic analysis methods, enabling users to upload sequences, perform analyses, and interpret results through interactive visualizations.

- **Analysing and Interpreting Genomics Datasets: [Genomics Training Repository]**

Tools: [Linux, Bash, HTML, RStudio, Nextflow concepts, NGS workflows]



- Built a practical genomics repository covering NGS concepts, genomics file formats, sequence quality control, mapping, and variant-calling workflows.
- Documented exposure to workflow management with Nextflow and viralrecon, strengthening preparation for scalable and reproducible bioinformatics pipelines.
- Connected Linux command-line analysis with RStudio-based visualization for interpreting genomic variant outputs.

- **Bioinformatics for Biologists: [Command-Line Bioinformatics]**

Tools: [Bash, Linux, Biological Data Workflows]



- Completed shell-based bioinformatics exercises focused on command-line workflows and biological data analysis foundations.
- Strengthened practical skills in reproducible directory organization, terminal-based analysis, and biological data handling.

- **Automated Data Extraction: [Scripting]**

April 2022

Tools: [Selenium, Pandas, Python]



- Implemented a web scraping script using Selenium and XPath to extract structured data from dynamic web pages.
- Processed and transformed scraped data using Pandas, ensuring clean and structured datasets for analysis.

PRESENTATIONS AND INVITED LECTURES

- **Keynote Address: Data-Driven Decision – Building the Future We Want with Statistics**

February 12, 2025

Ghana Association of Statistics Students (GASS), KNUST

- Delivered a keynote speech on the role of statistics in shaping data-driven decision-making.
- Highlighted the impact of statistical methods in policy development, business intelligence, and research.

- **Effective Teaching Strategies**

January 20, 2025

Department of Statistics and Actuarial Science, KNUST

- Delivered a structured presentation to newly admitted lecturers on effective teaching and assessment strategies.
- Explained the two main types of teaching strategies: teacher-centered and student-centered approaches.

- **Introduction to Git and GitHub for Research Collaboration**

November 19-21, 2024

Department of Statistics and Actuarial Science, KNUST

- Conducted a hands-on workshop for graduate and research assistants on version control using Git and GitHub.
- Explained key concepts such as repositories, branching, commits, pull requests, and merging.

COMPUTING SKILLS

- **Programming Languages:** Python, R, SQL
- **Web Technologies:** Flask, HTML, CSS, JavaScript, Bootstrap
- **DevOps & Version Control:** Git, GitHub, Conda, Bash scripting

CERTIFICATIONS

- **Basic Molecular Biology Module 3** August 2025
- **An Introduction to Linux, Bash Scripting, and R** March 2025
- **Full Stack Software Engineering** February 2025
- **Basic Molecular Biology Module 1** September 2024
- **Data Science with R** April 2024
- **Nanodegree Data Analysis** October 2022
- **Data Science Bootcamp** April 2022

REFERENCES

Available upon request.